

Bio-inspired Structural Engineering

Honeycomb Structures and Bone inspired Load bearing Structres

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Introduction

- A bee, and roughly 300 million years of vertebrate evolution, arrived at the same structural answer — independently.
- Honeycomb: built by bees to store the most honey using the least wax.
- Trabecular bone: grown by the body to withstand the most load using the least material.
- Neither was "trying" to optimize in an engineering sense — yet both converge on the same underlying rule: **put material only where the load actually needs it.**
- This project asks: what is that rule, why does it work, and how are humans now deliberately re-engineering it — through geometry, biology, computation, and manufacturing?
- We'll walk through all four lenses, then demonstrate the principle live with an interactive structural simulation we built ourselves.

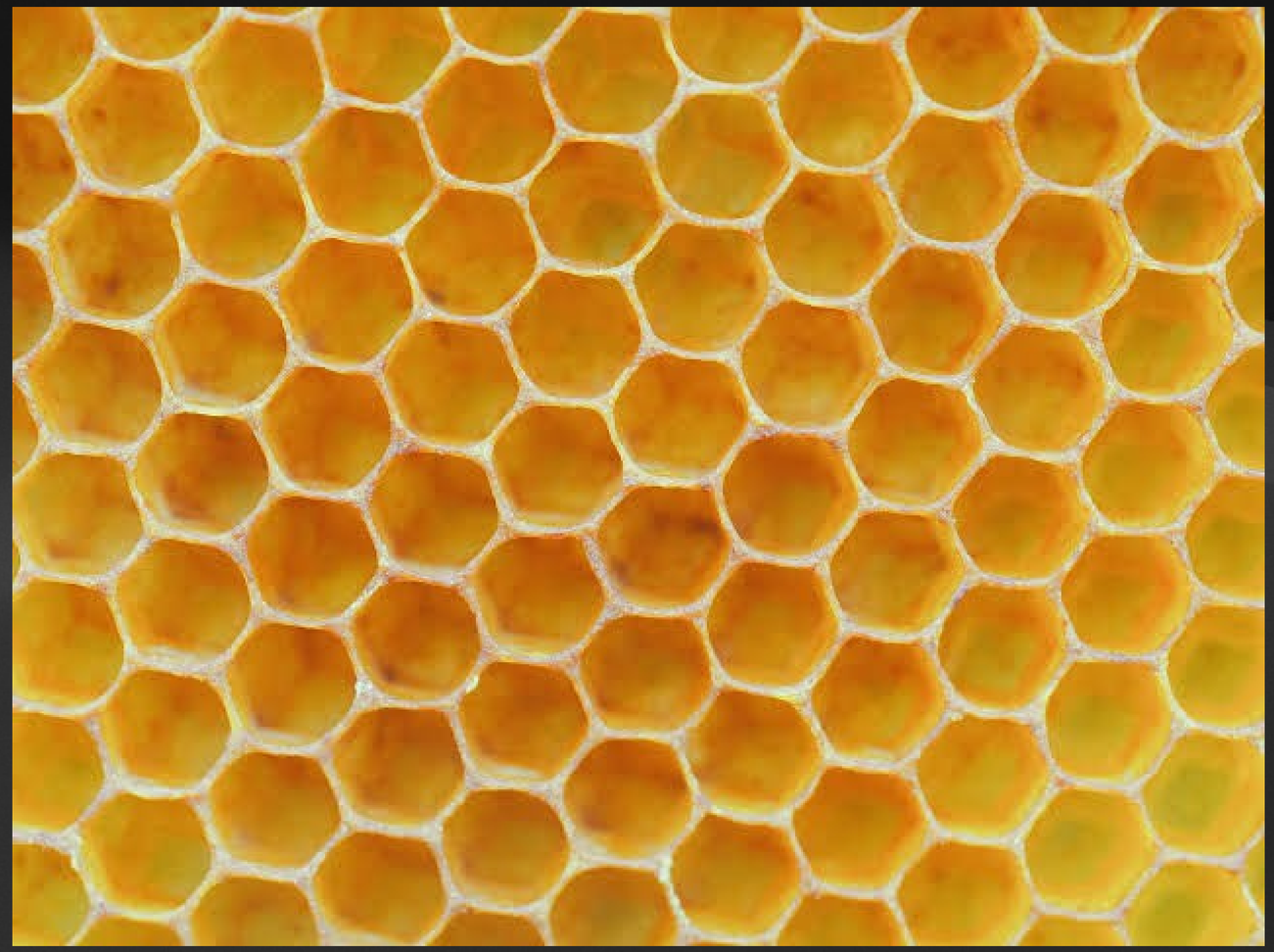


Honeycomb biology, hexagonal geometry, and honeycomb applications

Paarth Shanbhag

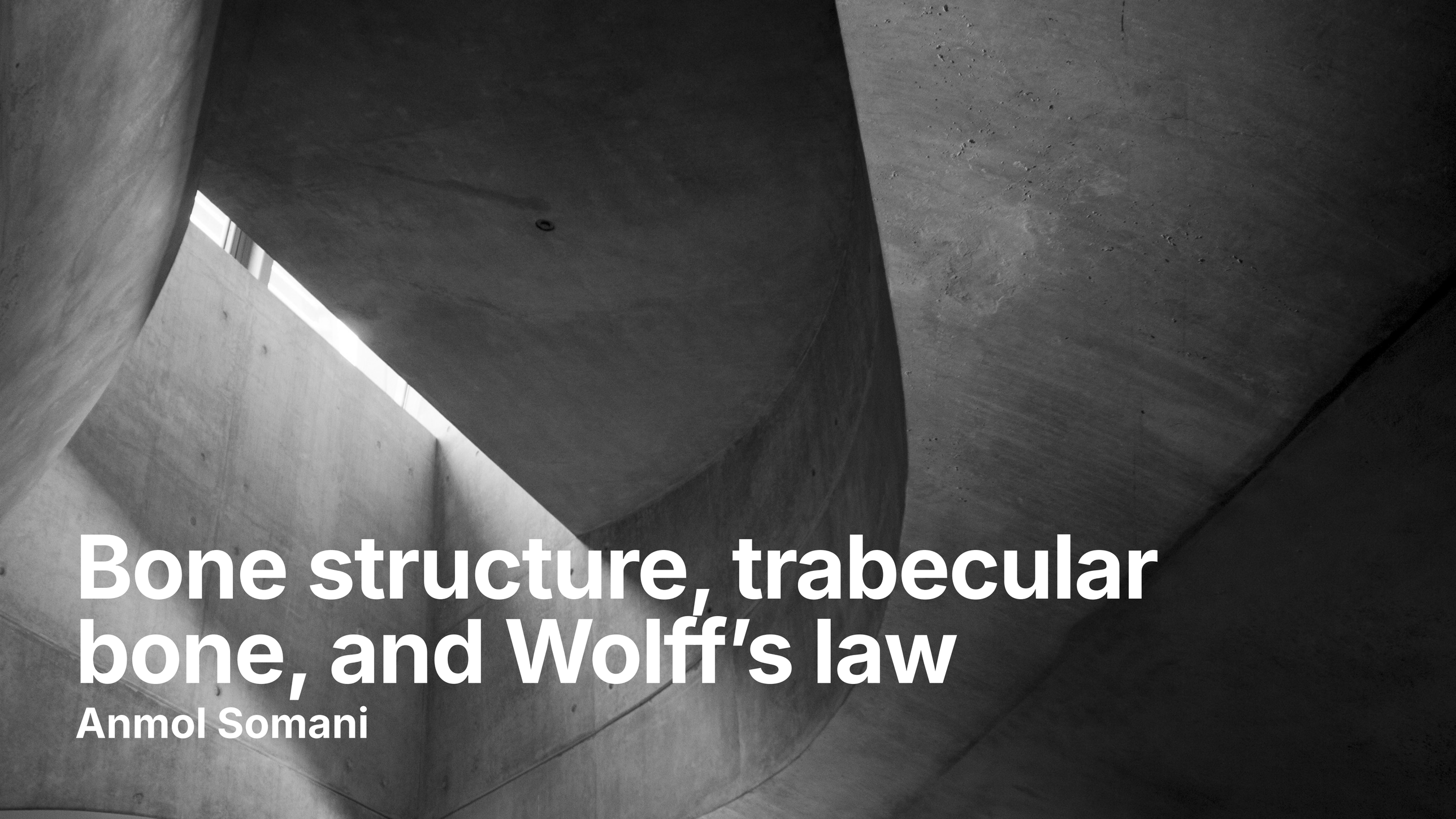
Honeycomb Biology

- Honeycombs are built by honeybees using wax.
- Bees use hexagonal cells to store honey, pollen, and raise young.
- The hexagonal pattern provides maximum storage with minimum wax/material.
- Hexagons fit together without leaving gaps, making the structure compact and efficient.
- This is an example of biomimicry: humans study nature and copy its efficient designs.



Hexagonal Geometry

- A hexagon has 6 equal sides and 6 angles.
- Regular hexagon: each interior angle = 120°.
- Hexagons can tile a surface completely without gaps or overlaps.
- Compared with many other shapes, the honeycomb pattern gives an excellent strength-to-weight ratio.
- The geometry distributes forces across the connected cells, making the structure strong and lightweight.



Bone structure, trabecular bone, and Wolff's law

Anmol Somani

The Architecture of Human Bone

The Living Structure

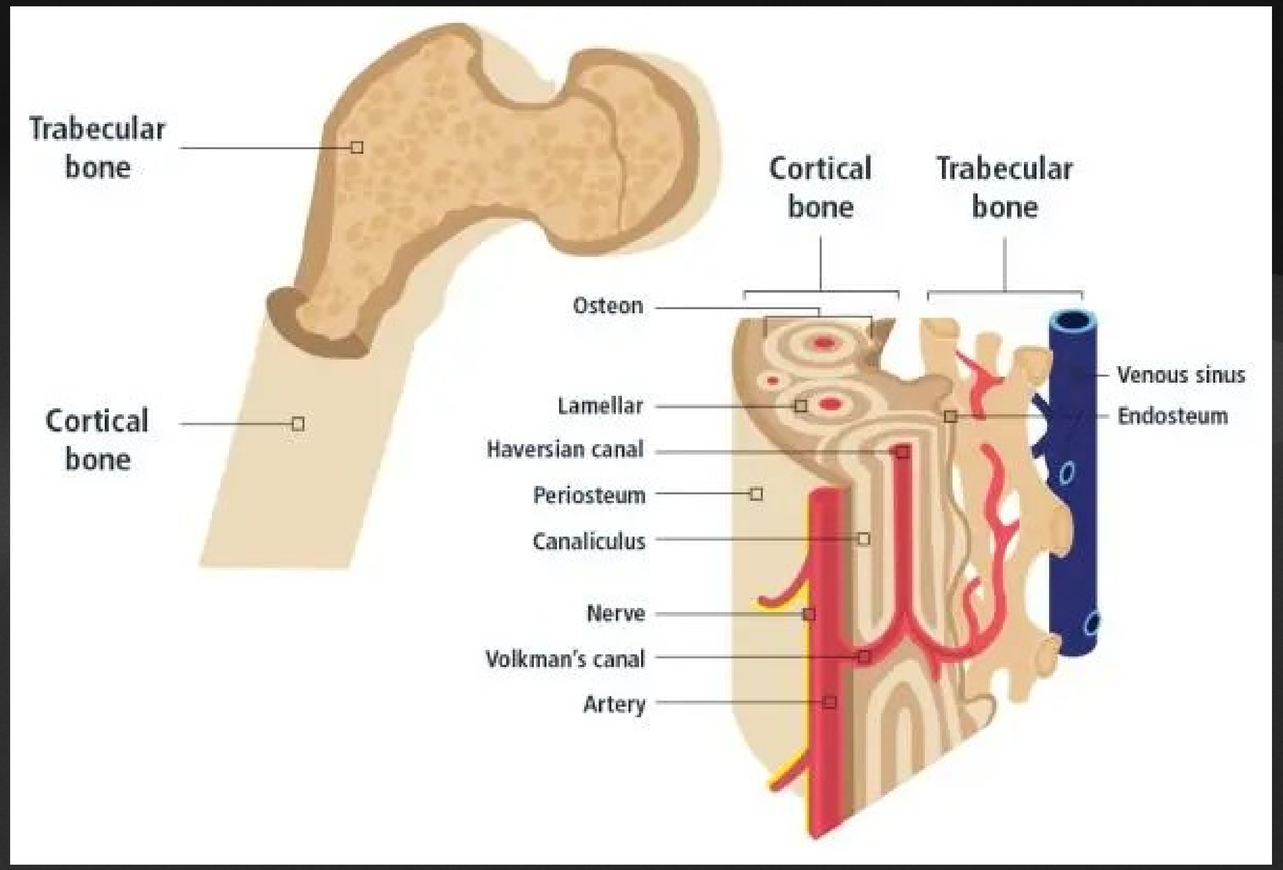
- Bones are living tissues that contain their own blood vessels, cells, proteins, and minerals.
- Because bone is living tissue, it has the unique ability to continuously grow, transform, and repair itself over the course of a lifetime.

The Dense Outer Shell (Cortical Bone)

- Human long bones are structurally tubular in shape.
- This tubular design is wrapped in a rigid, strong outer shell known as the cortical layer.
- This dense outer layer exists to provide primary structural support for the body and act as a protective armor for vital organs.

The Spongy Inner Mesh (Trabecular Bone)

- Surrounded by the hard cortical shell is a much lighter, spongier core called trabecular bone.
- Healthy trabecular bone is built from complex, plate-like internal structures rather than being completely solid.
- These specific plate-like formations provide the skeleton with superior mechanical strength.
- This spongy internal architecture gives the bone essential resistance against bending and compression when placed under heavy loads.



Wolff’s Law

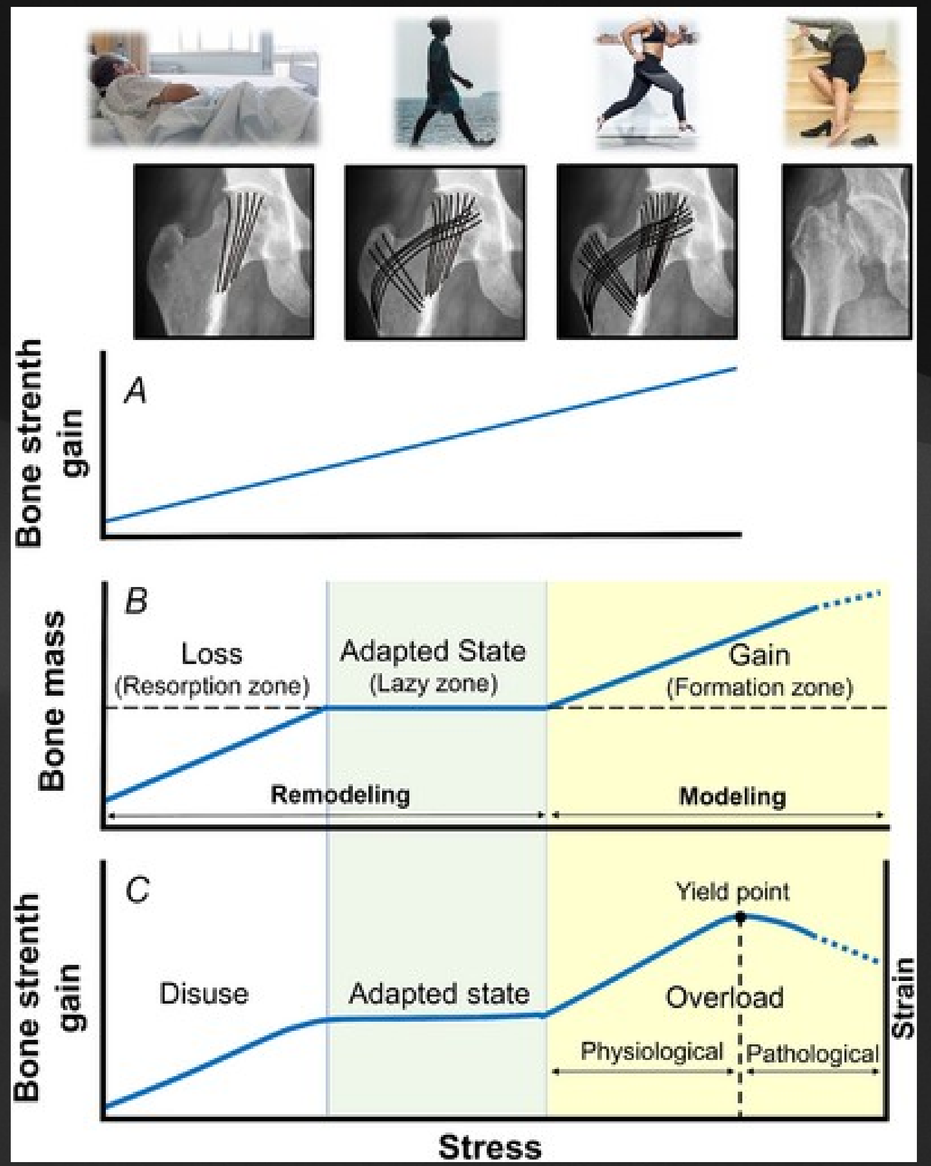
The Origin & Core Principle

- Proposed by German anatomist Julius Wolff in 1892
- Bone is a dynamic organ that continuously adapts its structure in response to repeated mechanical loads.
- **Increased Loading:** Triggers an adaptive response where bone remodels and strengthens to resist applied forces.
- **Decreased Loading:** Shifts cellular homeostasis toward a catabolic state, causing bone to resorb.

How Bone Senses Force (Mechanotransduction)

- Mechanical signals are converted into biochemical cellular responses through four sequential steps:

1. **Mechanocoupling:** Physical force deforms the bone tissue.
2. **Biochemical Coupling:** Deformation triggers chemical signaling molecules.
3. **Signal Transmission:** Signals spread across bone cells.
4. **Cell Response:** Osteoblasts and osteoclasts alter bone shape and density.

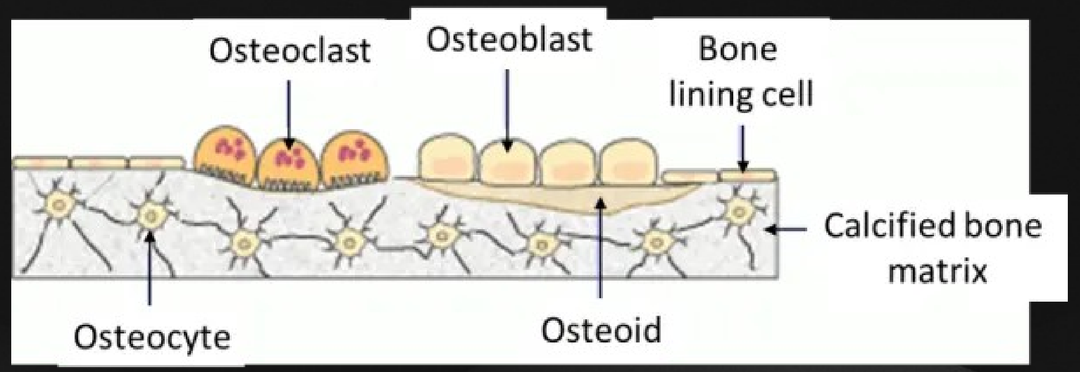


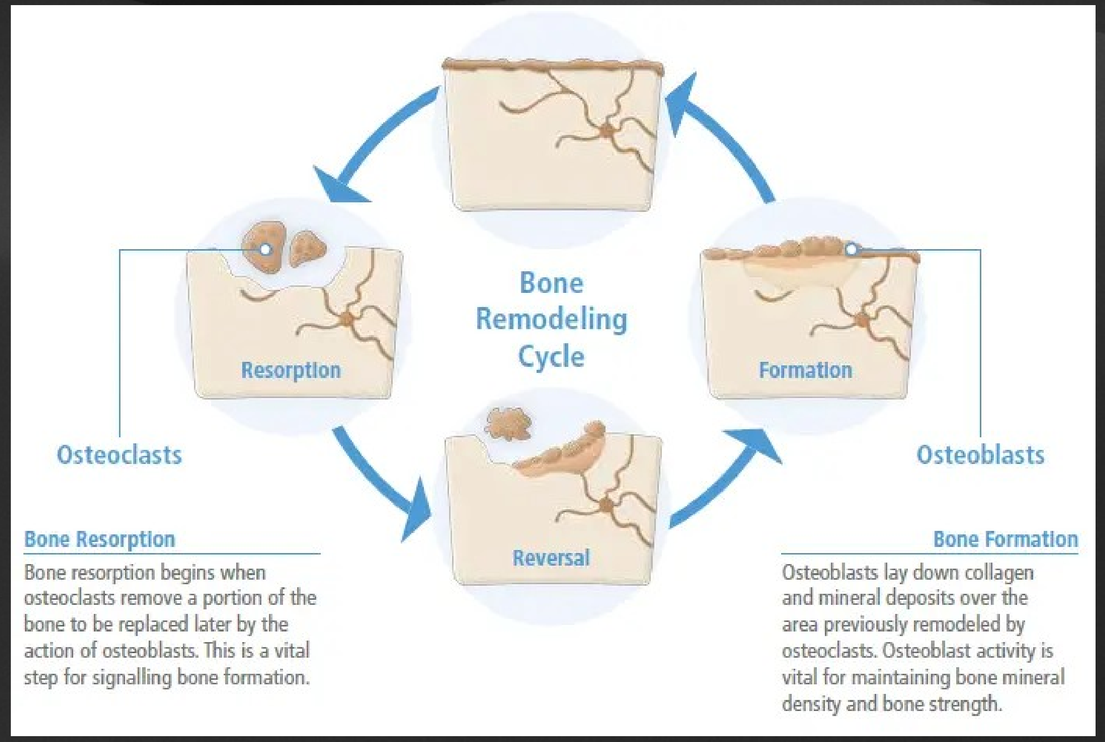
Real-World Triggers of Bone Remodeling

- **Gait & Biomechanics:** Walking forces (knee adduction moment) directly correlate with an expanded subchondral bone cross-sectional area.
- **Increased Weight (Obesity):** Higher BMI places excessive mechanical load on joints, driving local tibial bone surface area expansion.
- **Joint Damage:** ACL tears and meniscal injuries disrupt shock absorption, transferring higher load to subchondral bone and triggering rapid remodeling.

The Adaptive vs. Maladaptive Paradox

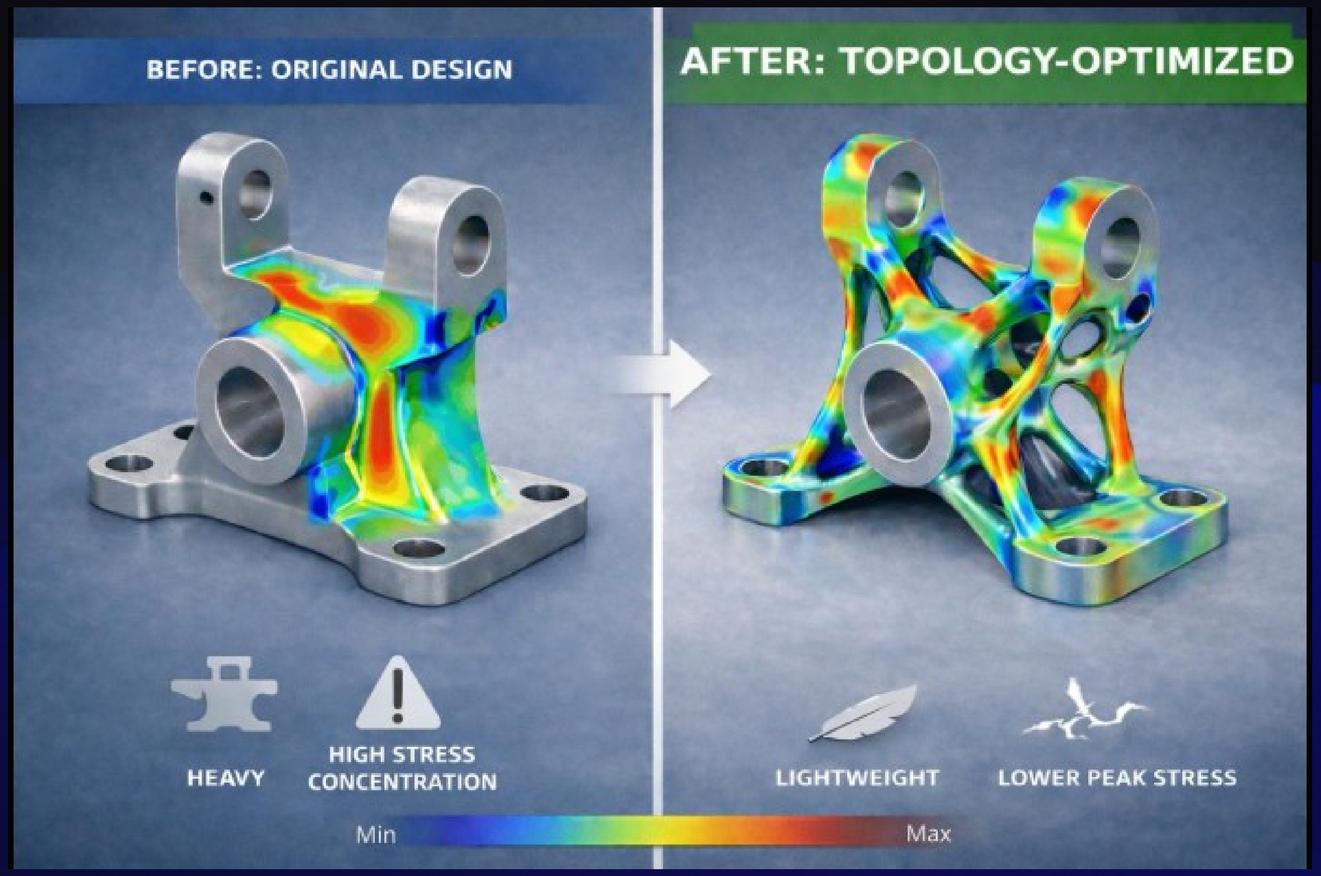
- **The Bone's Goal:** Broadening bone surface area and remodeling trabeculae is a healthy, adaptive response to spread out heavy joint loads.
- **The Untoward Consequence:** As subchondral bone expands and hardens under Wolff's Law, it alters joint mechanics and accelerates cartilage loss, triggering early knee osteoarthritis.





Computational modelling, topology optimisation

Kavya Chaturvedi



FROM BONE TO COMPUTER MODEL

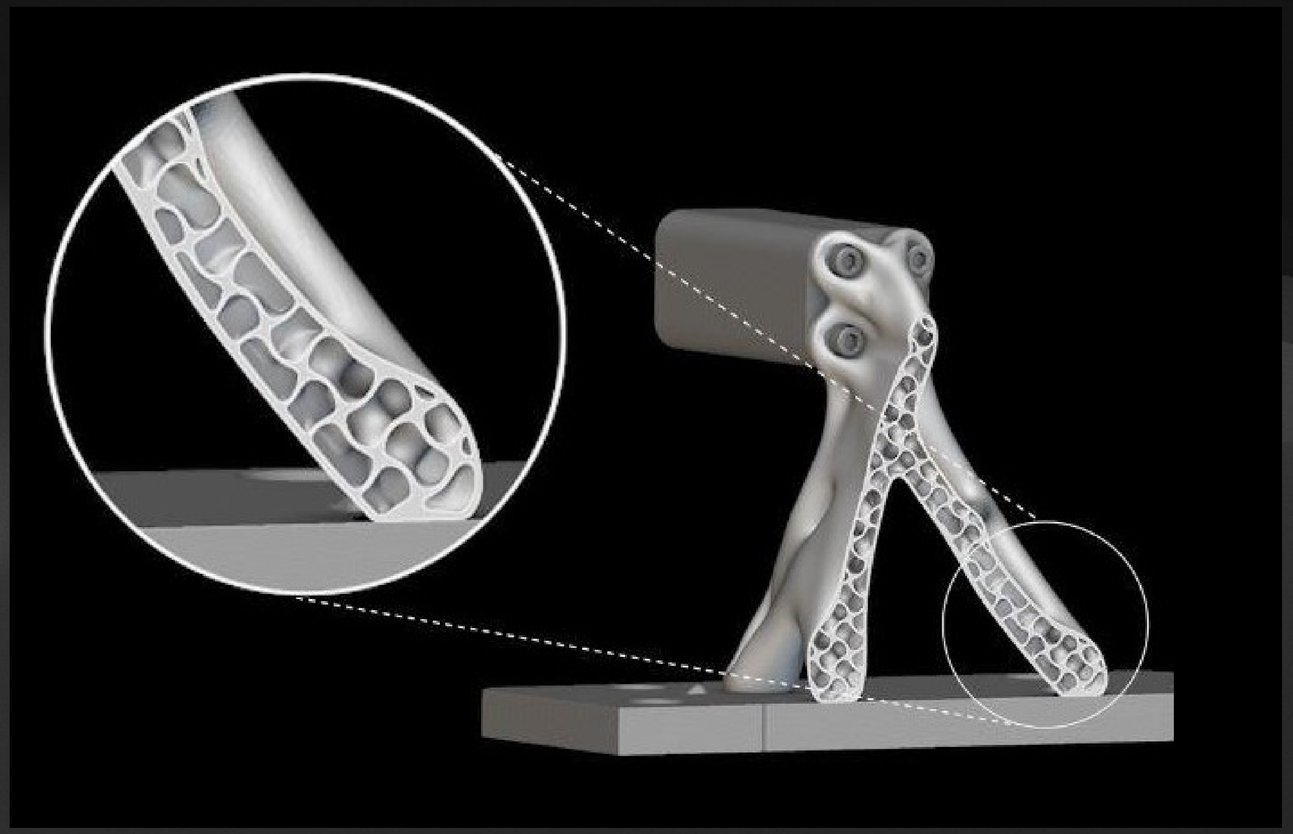
How Nature Inspires Structural Design

Trabecular Bone

- Many bones contain a porous internal network called **trabecular bone**.
- Its architecture is related to the **mechanical environment and loading** experienced by the bone.
- Bone remodelling adapts its internal architecture to mechanical demands, producing an efficient load-bearing structure.

Wolff’s Law → Engineering Principle

Bone remodelling is associated with the mechanical environment: its internal trabecular architecture adapts to loading, producing a mechanically efficient structure.



Jang & Kim (2008)

- Studied the trabecular architecture of the **human proximal femur**.
- Used finite-element modelling and topology optimisation.
- Their simulations produced trabecular patterns with **strong similarities to observed bone architecture**.

Engineering Translation

Nature: Mechanical loading to adapted bone architecture

Engineering: Mechanical loading to computational optimisation to efficient structure

Engineers do not copy every detail of bone biology. They translate the mechanical principle into a computational design process.

- Reference material - https://www.sciencedirect.com/science/article/abs/pii/S0021929008002801

FEA to TOPOLOGY OPTIMISATION

How the Computer Determines Where Material Is Needed

CAD MODEL

The engineer creates a digital model and defines:

- Geometry / design domain
- Material properties
- Applied loads
- Boundary conditions
- Structural constraints
- Optimisation objectives

FEA — FINITE ELEMENT ANALYSIS

The structure is divided into many small **finite elements**.

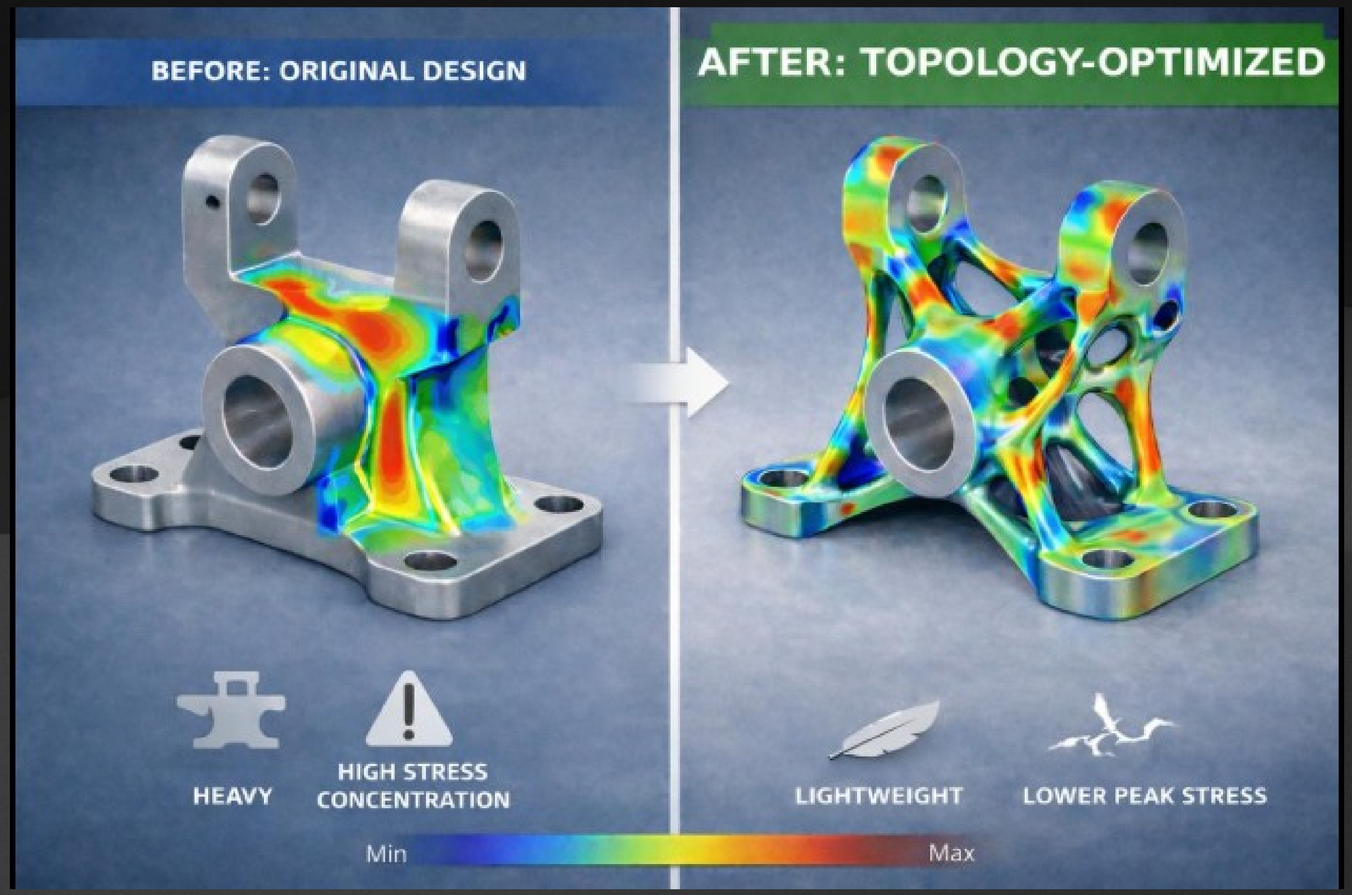
FEA calculates the structural response, including:

Stress • Strain • Displacement • Deformation • Strain Energy

STRESS MAP

A stress map is a visual representation of how stress is distributed throughout a structure. It allows engineers to identify regions experiencing relatively high or low mechanical loading.

This is useful because engineers can see where the structure is heavily loaded and where material may be contributing less to the structural response.



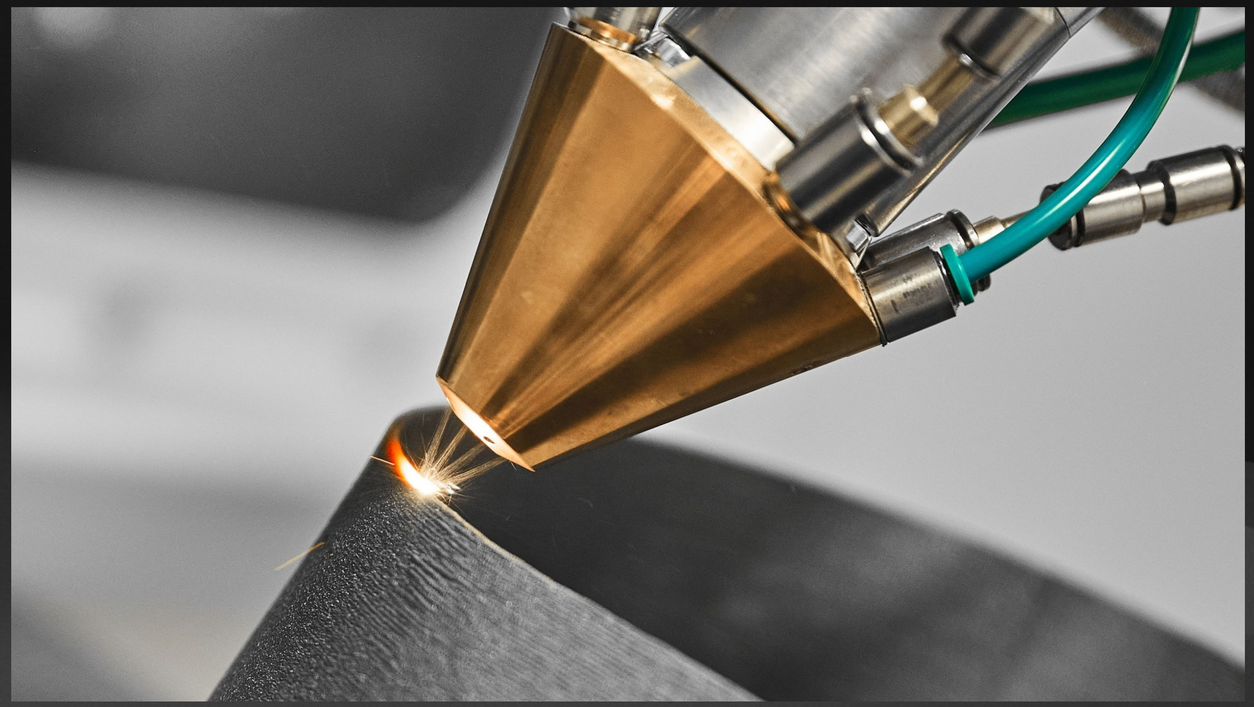
FEA asks:
“What happens to this design when I apply the loads?”
Topology optimisation asks:
“How should the material be distributed so the structure meets the required performance?”



Manufacturing, industrial applications, advantages, limitations, and final connection

Bhavyakumar Soni

Manufacturing — From Idea to Object



Machining can't reach inside a closed lattice — no way in for a drill

3D printing builds layer-by-layer, inside-out

Honeycomb: corrugate → expand → bond into panels

Trabecular: generative design → layer fusion → post-process



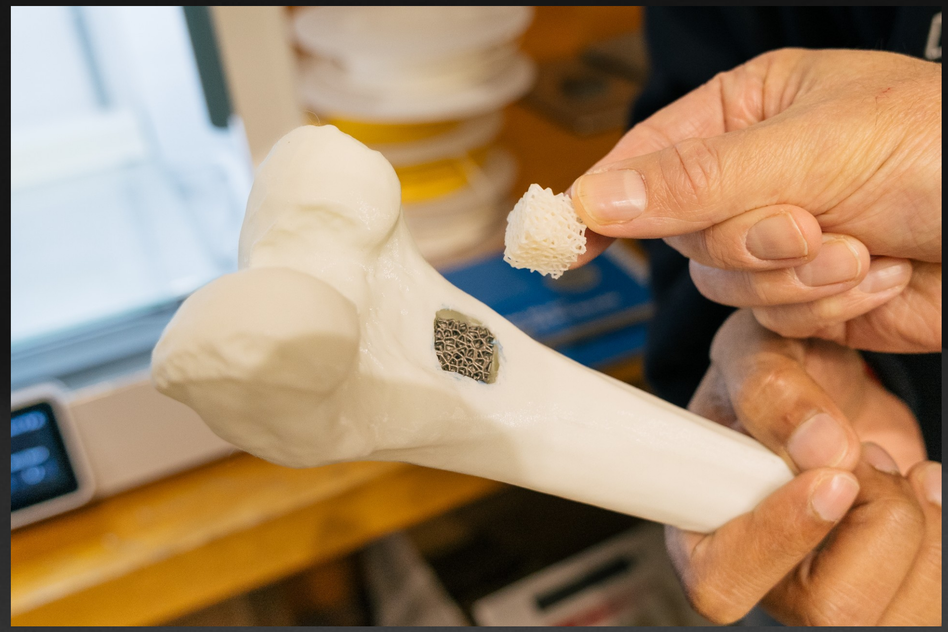
Where It's Already Working

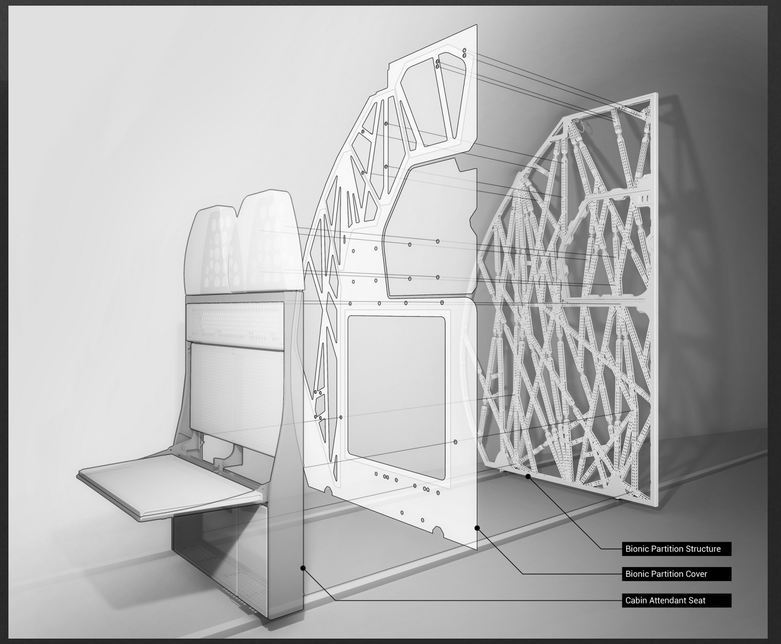
Honeycomb: aircraft panels, crash structures, packaging

Airbus: bone/honeycomb-inspired walls cut weight ~50%

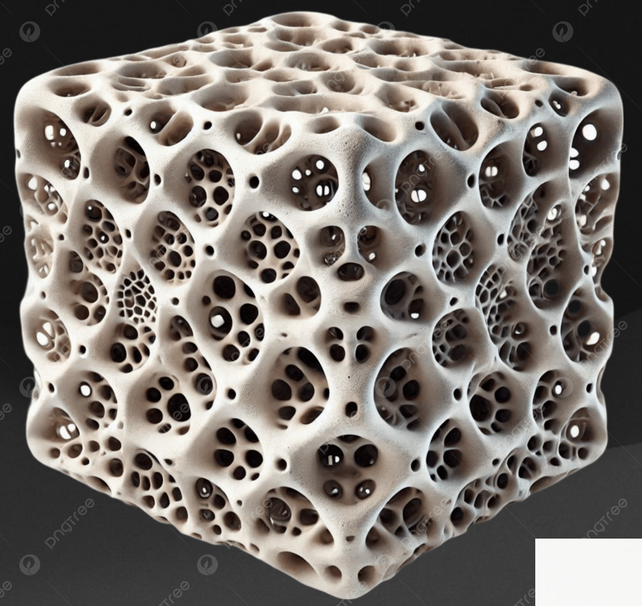
Trabecular-inspired: titanium implants that bond with real bone

Also: aerospace brackets, dental implants, footwear





Advantages, Limits & the Bigger Picture



Weight cut without losing strength — material only where load needs it

Limitation: 3D printing still slow, costly — not yet mass-market

Nature runs this biologically (Wolff's Law); engineering runs it computationally now — same rule, different hardware



Honeycomb Structures: Pros & Cons

Advantages

- **Exceptional stiffness-to-weight** — a thin core between two skins acts like a solid beam at a fraction of the mass
- **Strong crush energy absorption** — cells collapse progressively, ideal for crash boxes and packaging
- **Material-efficient geometry** — hexagons tile with no gaps, using the least wall material per cell
- **Mature, low-cost production** — expansion and corrugation methods scale to aluminium, Nomex, paper and polymers
- **Insulation bonus** — trapped air in the cells damps heat and sound

Limitations

- **Anisotropic** — very strong out-of-plane, much weaker in-plane and in shear
- **Skin–core debonding** — sandwich panels can delaminate, and point loads dent the core
- **Moisture traps** — water entering sealed cells can corrode metal cores
- **Hard to inspect and repair** — hidden internal damage needs specialised NDT
- **Uniform, not adaptive** — identical cells everywhere, regardless of where loads flow

Bone-Inspired Structures: Pros & Cons

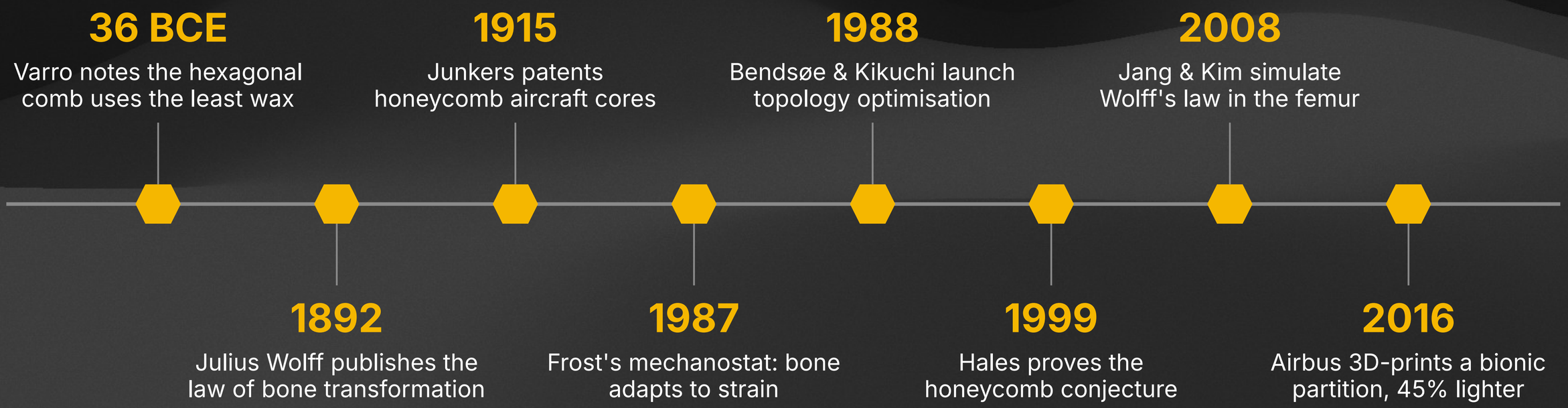
+ Advantages

- **Material along load paths** — density is graded to match stress, so mass goes only where it is needed
- **Handles multi-directional loads** — 3D trabecular-style lattices resist compression, bending and torsion
- **Damage tolerant** — redundant struts let loads reroute when one member fails
- **Tunable stiffness and porosity** — implants can match bone stiffness, reducing stress shielding
- **Promotes bone ingrowth** — porous titanium surfaces help implants fuse with living bone

− Limitations

- **Needs additive manufacturing** — 3D printing is slow, costly and limited in part size
- **Fatigue-sensitive** — printing defects and rough thin struts can start cracks
- **Heavy computation** — FEA plus topology optimisation demands time and expertise
- **Hard to inspect** — internal lattices often need CT scanning to verify
- **No self-repair** — unlike real bone, engineered parts cannot remodel or heal

Research Timeline



References

[1] Wolff, J. (1892). *Das Gesetz der Transformation der Knochen*. Berlin: A. Hirschwald.

[2] Frost, H. M. (1987). Bone “mass” and the “mechanostat”: A proposal. *The Anatomical Record, 219*(1), 1–9.

[3] Bendsøe, M. P., & Kikuchi, N. (1988). Generating optimal topologies in structural design using a homogenization method. *Computer Methods in Applied Mechanics and Engineering, 71*(2), 197–224.

[4] Gibson, L. J., & Ashby, M. F. (1997). *Cellular Solids: Structure and Properties* (2nd ed.). Cambridge University Press.

[5] Hales, T. C. (2001). The honeycomb conjecture. *Discrete & Computational Geometry, 25*, 1–22. https://doi.org/10.1007/s004540010071

[6] Jang, I. G., & Kim, I. Y. (2008). Computational study of Wolff’s law with trabecular architecture in the human proximal femur using topology optimization. *Journal of Biomechanics, 41*(11), 2353–2361. https://doi.org/10.1016/j.jbiomech.2008.05.037

[7] Airbus. (2016, March 3). *Pioneering bionic 3D printing*. https://www.airbus.com/en/newsroom/news/2016-03-pioneering-bionic-3d-printing

Thank You